Manuscript outline

Cell enlargement, polarity competition and migration persistence

Vision-aligned manuscript outline with proposed main figures and detailed legends

19 October 2026 · Figures 2–4 revised against the 19 October program

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations supply empirical motivation; new paired WGD motility results retain a substantial, explicitly reserved contribution.

Figures 2–4 focus on three questions: which measurements separate explanations for difficult polarity replacement; how local activity becomes global polarity on a ring; and how physical site motion, pattern organization and geometry contribute to angular memory. Retained simulations and targeted distance–duration and Spring-ring experiments supply the evidence. Detailed assay maps and secondary diagnostics are supporting material.

This is a working manuscript outline. Figures 1 and 5 and the reserved Figure 6 experimental section retain their preceding content. Activity polarity, modelled replacement, observed migration and conditional OU velocity memory remain distinct readouts; their biological connection is an experimental hypothesis.

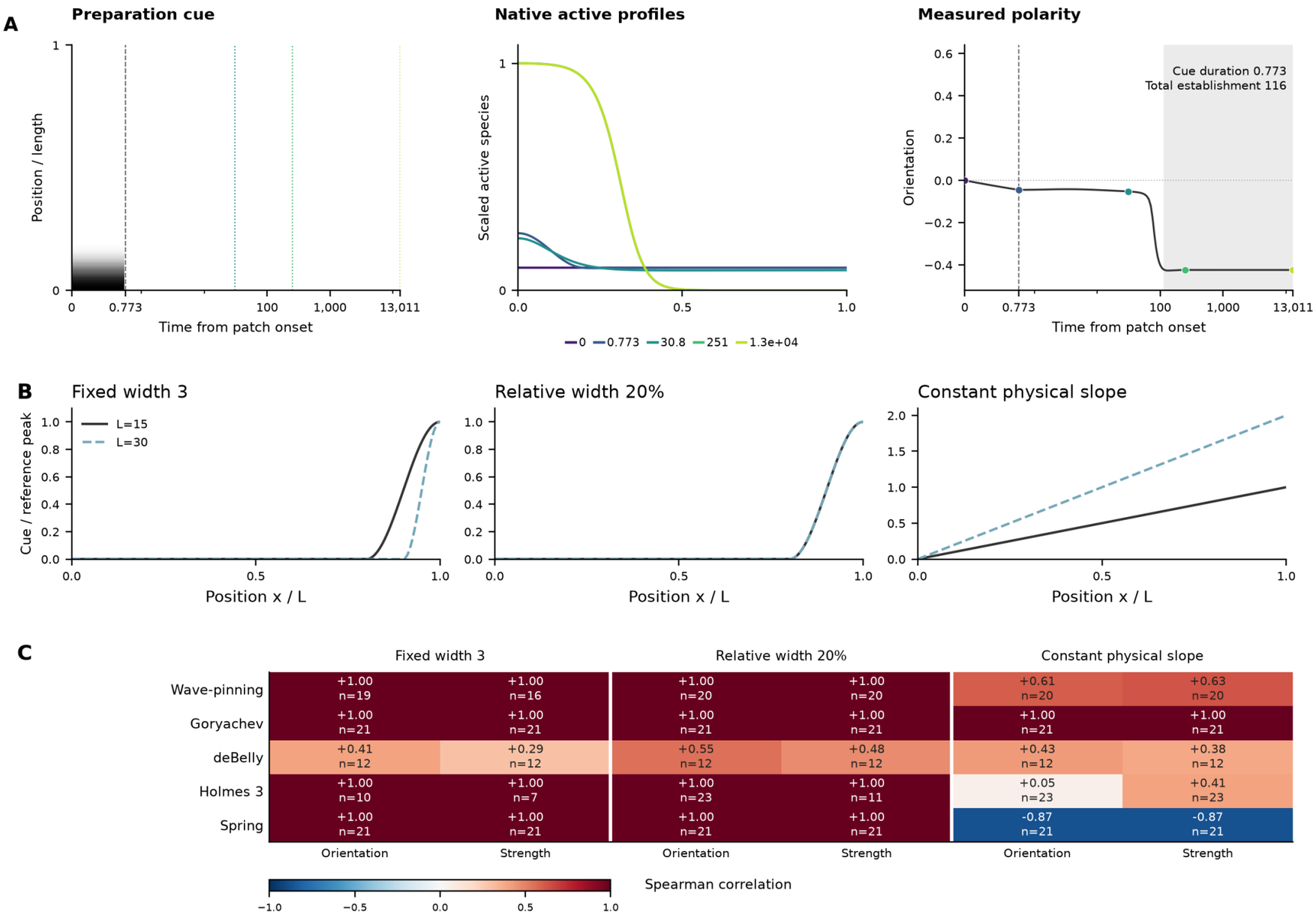
Proposed main figures

- Figure 1. Enlargement changes the cue duration required for lasting polarity replacement
- Figure 2. Similar reversal trends, distinct challenge responses
- Figure 3. Geometry, cue placement and duration shape retained polarity
- Figure 4 · Directional memory and physical site dynamics
- Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Manuscript outline · continued

- Figure 6. RESERVED: new WGD motility-assay results

Figure 1. Five models and three finite challenges



Current canonical comparison, 10 October Spearman revision; corrected independent-endpoint deBelly.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

Comparative theoretical foundation: five established models, three finite challenges, and two distinct scalar replacement gates.

Detailed legend

A, Example wave-pinning preparation from a homogeneous state using a finite left patch. The cue ends at 0.773 native time units, whereas the subsequent basal relaxation establishes the polarized profile later (reported establishment arrival about 116). Coloured profiles and orientation samples distinguish input duration from autonomous development. The grey late interval is verification, not additional stimulation. Profiles are scaled for display.

B, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

C, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

deBelly uses independently commanded corrected endpoints, rather than the deprecated in-loop centered extension. Spring length and time retain native units and are not silently converted to micrometres or seconds. Model-specific axes and stimulus definitions prevent interpreting cross-model amplitudes or clocks as biologically calibrated equivalents.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

Narrative points and conclusions

- Fixed-width and relative-width patches show perfectly increasing rank order among observed successes in Wave-pinning, Goryachev, Holmes 3 and Spring; enlargement often increases the finite stimulation required for a retained response.
- The response depends on the cue geometry and model. Spring has a negative constant-slope association (Spearman correlation -0.874 for both gates), while Holmes orientation under the constant-slope cue is nearly unassociated with length (0.047).
- Corrected deBelly curves are nonmonotone despite moderate positive rank correlations. The full duration curves and eligibility ranges are essential to the result.
- These are deterministic polarity-replacement experiments. They motivate, but do not measure, spontaneous polarity memory or migration persistence.

Proposed measurements and experimental follow-up

Mechanism-linked cellular follow-up would match local input support, amplitude and duration, track the original and challenged activity regions through stimulation and withdrawal, and distinguish orientation replacement from recovered activity strength. This validation is proposed and has no experimental results in this outline.

Scope, limitations and pending validation

Correlations are conditional on observed scalar success, rather than an unbiased summary of all possible sizes or a universal scaling law. Ineligible starts are distinct from responses unobserved by the finite horizon.

A retained scalar response can coexist with a changing native field. Only 13 of the 72 corrected deBelly scalar successes pass the separate native-field tail flag. Resolution checks from the earlier centered-deBelly campaign cannot qualify the corrected replacement thresholds.

Sources and reproducibility

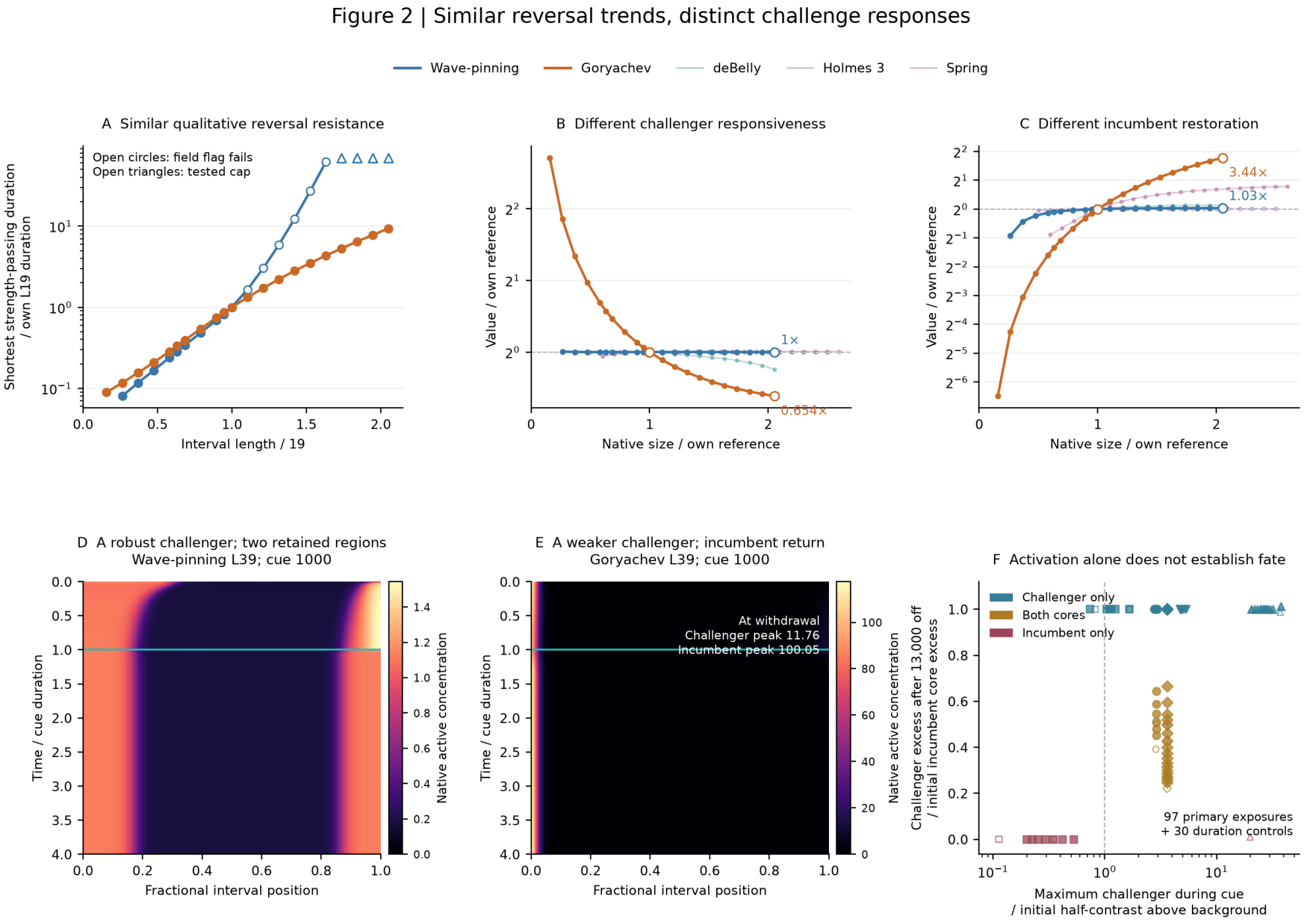
Three-test protocol and Spearman rule

Correlation table

Corrected endpoint method

Current duration curves

Figure 2. Similar reversal trends, distinct challenge responses



A: fixed-support scalar-strength thresholds; similar means increasing over observed successes, not equal slopes or calibrated times.
B–C: all 97 basals; WP/Goryachev highlighted. C is a specified local active-loss response, not general stability.
D–E: exact native fields, model-specific zero-based scales; cyan = withdrawal. F: final fixed-core geography; filled = primary, open = duration control.

Figure 2. Similar reversal trends, distinct challenge responses

Figure 2. Similar reversal trends, distinct challenge responses

Similar size-dependent reversal behavior can arise through distinct mechanisms, which predict different observable responses to polarity challenges.

Detailed legend

Similar size-dependent reversal behavior can arise through distinct mechanisms, which predict different observable responses to polarity challenges. Wave-pinning and Goryachev provide the lead comparison: both require longer fixed-support stimulation at increasing size over their observed scalar-strength successes, while their gain and local-restoration responses differ. Similarity is qualitative rather than equality of slopes or calibrated physical durations.

A, Shortest tested fixed-support pulse duration passing the original scalar-strength replacement gate, divided by each model's own L19 duration (WP 73.5; Goryachev 377.5 native units). Both curves strictly increase over observed successes: WP L5–31, 5.906–4540; Goryachev L3–39, 33.726–3540. Open triangles retain WP L33–39 censoring at the 5120-unit tested cap; no time is assigned beyond that cap. Open circles retain scalar successes failing the separate native-field tail flag (WP L21–31). Preparations are unavailable at WP L1/L3 and Goryachev L1. Strength requires reversal of the signed scalar polarity with magnitude within 1% of its initial value, in addition to the original orientation, contrast and final-window persistence conditions; this does not prove complete field equilibration.

B, Calculated cue-weighted incremental primary-activity production per native opposing input, for all 97 qualified basals. The fixed endpoint raised-cosine support is 3, or 0.2 in Spring. Gain and size divide by each model's own reference (L19, Holmes L20, Spring $L_0 1$). Goryachev's near-doubling gain is 0.654 of the reference; lower available substrate supports weaker initial activation under the same local input. This calculation is conditional on the native input channel.

C, Sham-relative deficit restoration after 0.5% of a smooth frozen-incumbent excess profile is deactivated; removed material enters an explicit inactive pool where present. Corrected deBelly changes Rac alone and initially preserves MCA/mechanics. All native fields evolve. The repaired fraction at one fixed native clock per model is divided by its own reference. Clocks are WP/Holmes 0.177827941, Goryachev 0.005623413, deBelly 5 and Spring 0.017782794; 0.1%/1% amplitude and numerical checks qualify all 97 cases. This measures response to a declared deficit, not general stability.

D–E, Exact native WP/Goryachev L39 fields during a 1000-unit fixed-support cue and the next 3000 units; cyan marks withdrawal. WP develops a robust challenger and retains two threshold-defined regions through the complete 13,000-unit off-window. Goryachev develops a smaller but nonzero challenger, below the original half-contrast, then restores its incumbent. Native concentration scales are model-specific and start at zero; these are different within-model native doses, not calibrated equal cross-model inputs.

Figure 2. Similar reversal trends, distinct challenge responses · continued

F, Native histories reanalysed for every primary exposure (97 filled points) and every prespecified half/double-duration anchor (30 open points). Horizontal coordinate is the largest challenger-core peak during the cue, above frozen basal background and divided by initial half-contrast. Vertical coordinate is challenger-core excess after 13,000 cue-free units divided by initial incumbent excess. Colour gives final original/mirrored-core geography: challenger only, both cores or incumbent only, using the same half-contrast threshold; it is not a tracked site identity or scalar replacement gate. Model symbols are WP circle, Goryachev square, deBelly triangle-up, Holmes diamond and Spring triangle-down. Among 89 primary exposures reaching the cue-on threshold, 23 retain both cores; a strong challenger is therefore insufficient to establish replacement. Across all 127 exposures, 30 threshold-positive cues have another final geography, and 2 challenger-only outcomes cross the threshold only after withdrawal. This quantifies observational ambiguity rather than identifying a unique molecular mechanism.

Cue peaks are WP 0.1, Goryachev 6.4, corrected deBelly 3.6, Holmes 1 and Spring 4; primary duration is 1000 native units, or 50 in Spring. Native times, channels and sizes are not calibrated across models. Thirteen preparation-ineligible sizes remain in the source table. The supporting WP mobility control passes retained orientation at a fixed 5120-unit pulse, while both native and fourfold-diffusion conditions fail A's stronger full-polarity-strength gate; fourfold diffusion acts throughout cue and recovery. Selected interventions and complete native profiles/histories are supporting evidence; no new integrations were run.

Narrative points and conclusions

- Wave-pinning and Goryachev both show increasing fixed-support scalar-strength duration over observed successes, but their magnitude, eligibility and censoring differ. This is qualitative similarity, not a shared scaling law.
- Incumbent material increases in every model, but concentration, width, initial responsiveness and local restoration change differently.
- Substrate availability and a declared local deficit assay jointly support a coupled activation/restoration explanation in Goryachev; their correlation does not partition reversal time.
- A strong challenger is insufficient: 23 of 89 activated primary exposures retain both original and challenger cores. Corrected deBelly39 exposes additional incumbent return after a shorter exposure.
- Transport and mechanical interventions show causal contributions under specified cues. The WP fixed-dose mobility control passes retained orientation while both conditions fail A's stronger strength gate; fourfold inactive diffusion acts throughout cue and recovery. Cue geometry changes intervention effects, so no single activity signature uniquely identifies an explanation.

Figure 2. Similar reversal trends, distinct challenge responses · continued

Proposed measurements and experimental follow-up

Measure incumbent peak, physical extent and integrated excess signal before stimulation, keeping concentration-like intensity distinct from calibrated amount.

Measure initial challenger activation under characterized fixed local input, and sham-relative incumbent deficit restoration under a specified small suppression.

Track original and challenger regions throughout stimulation and withdrawal; pair their local activity with substrate, attachment, flow or extension measurements when testing the corresponding mechanism.

Scope, limitations and pending validation

Native clocks and channels differ; the assays are not cross-model biological calibrations.

Positive restoration, strong activation and final fixed-core geography each leave observational equivalences; no unique molecular model, general stability or velocity law is inferred.

Deterministic sizes and doses are conditions rather than independent experimental replicates. Finite-window patterns do not prove attractors.

Sources and reproducibility

Evidence, methods and limits

Mechanism × observable matrix

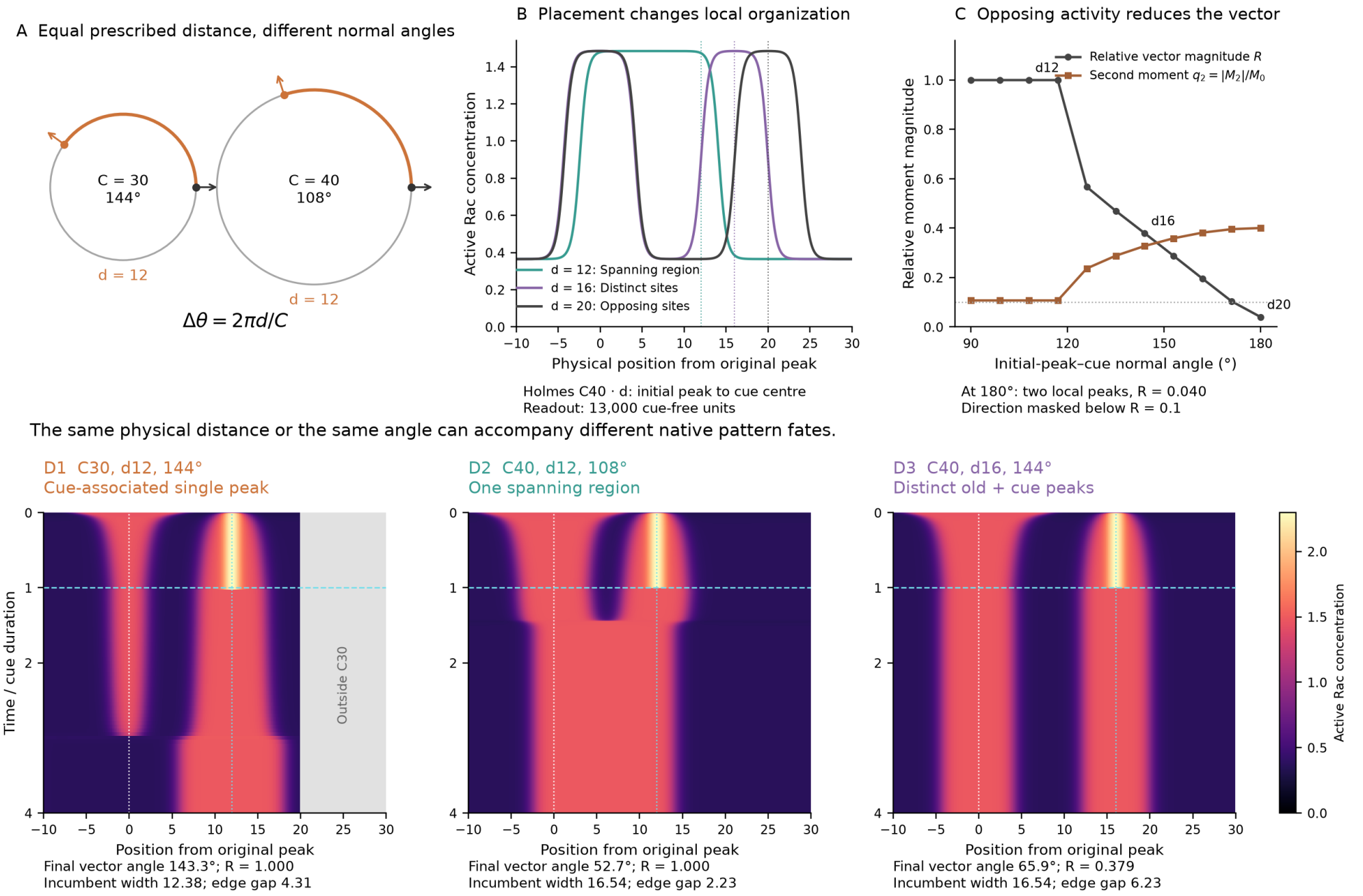
All 127 native activation/fate observations

Threshold and dose discrimination

Figure 3. Geometry and the activity vector · image 1 of 2

Figure 3 · Geometry connects local polarity organization to an activity vector

A larger perimeter converts the same physical separation into a smaller angular separation.



Native Holmes histories · cue 0–1000 (cyan withdrawal), fixed full support 3 and peak 1. Field display ends at 4000; final readout is 14000.
Activity vectors describe spatial activity, without assuming a cell-movement law. Complete maps and descriptor tests are linked in the evidence.

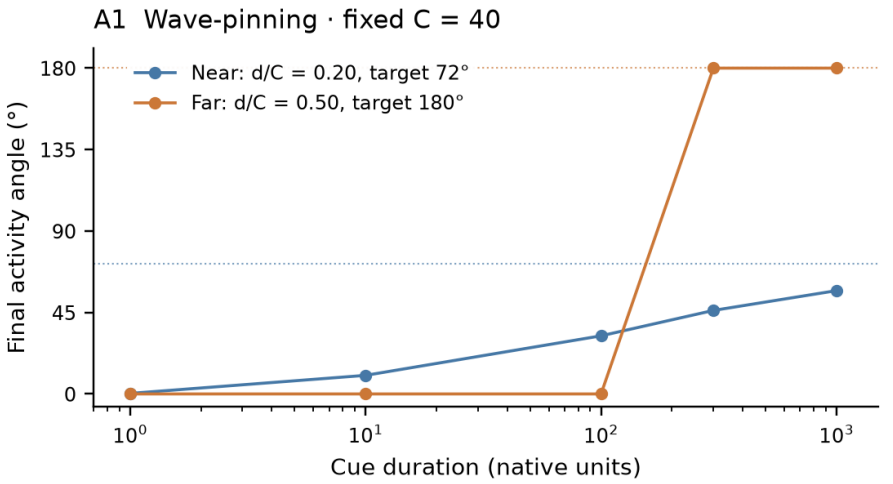
Revised compact geometry sheet; all plotted fields are retained native simulations.

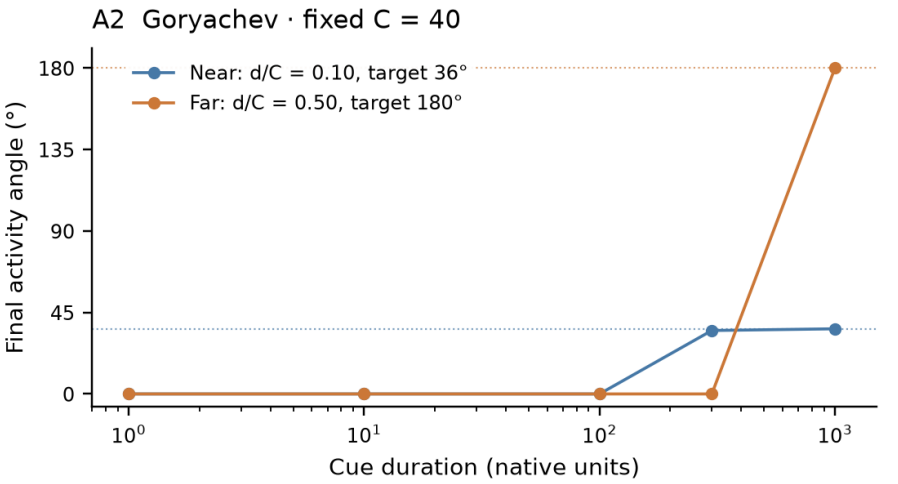
Figure 3 continued. Distance and finite cue duration · image 2 of 2

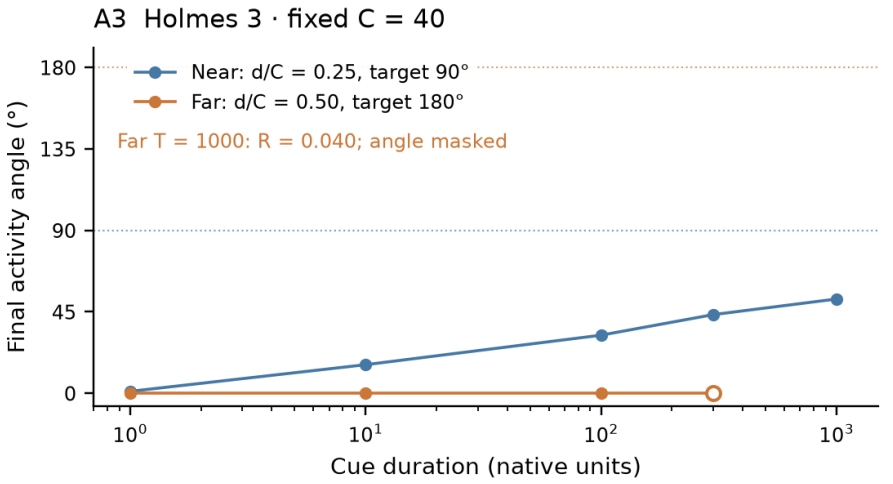
Figure 3 continued · Distance and finite cue duration alter retained activity orientation

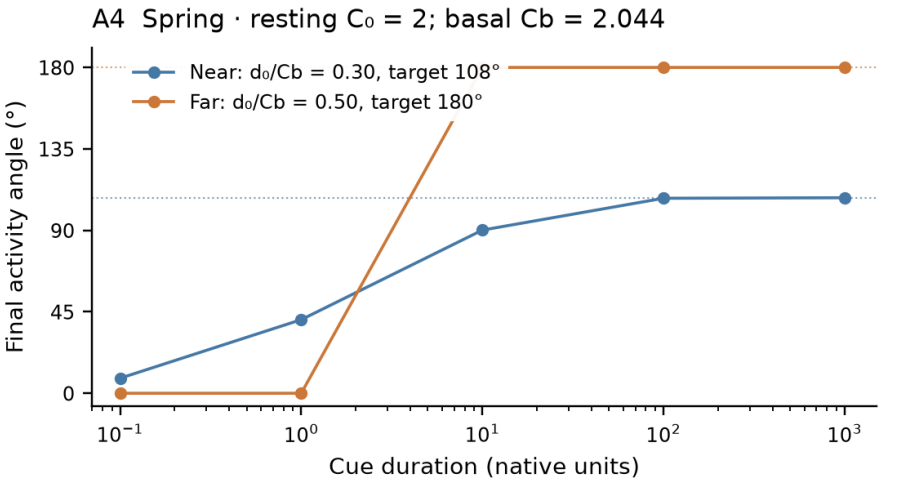
All points follow 13,000 cue-free units. Curves connect tested durations; dotted lines mark prescribed cue normals.

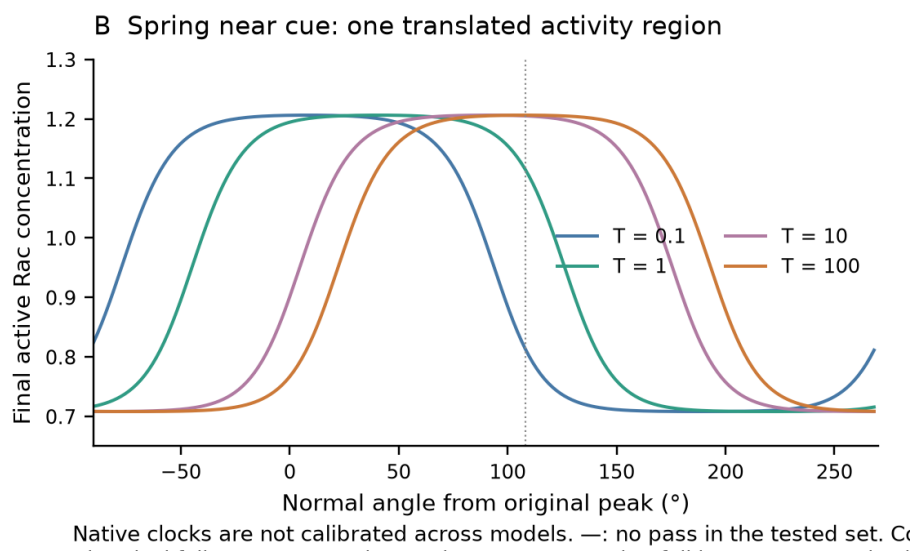
Open circles: failed separate field-tail gate. The full-field tail was not remeasured for reused chemical T = 1000 references.











C Shortest tested passes depend on the endpoint

Same cases	Cue ±15°	Core ≤½	R at T1000
Wave-pinning near	1000	—	1.000
Wave-pinning far	300	300	1.000
Goryachev near	300	300	1.000
Goryachev far	1000	1000	1.000
Holmes 3 near	—	—	1.000
Holmes 3 far	—	—	0.040
Spring near	100	100	1.000
Spring far	10	10	1.000

Native clocks are not calibrated across models. —: no pass in the tested set. Core loss is a fixed-region activity readout, not site ancestry.
Chemical full support 3, native peaks 0.1/6.4/1; Spring full instantaneous physical support 0.2, peak 4. Full maps, tail flags and threshold sensitivities remain linked.

Compact four-model duration-response continuation with native Spring profile evidence.

Figure 3. Geometry, cue placement and duration shape retained polarity

Closed geometry connects local activity to global direction and magnitude; distance and duration alter retained orientations and regional activity.

Detailed legend

A, Circular boundaries drawn at a common physical scale. A prescribed shorter-arc separation d of 12 native length units from the initial incumbent peak to the challenger cue centre gives an outward-normal difference of 144° at circumference 30 and 108° at circumference 40. B, Native Holmes Rac profiles 13,000 units after withdrawal at C40, from the identical qualified basal state, with a raised-cosine challenger of full support 3, native peak 1 and duration 1000. Increasing prescribed cue-centre separation changes a connected spanning region (d12) to distinct old- and cue-associated peaks (d16 and d20). Dotted vertical lines mark challenged positions. C, Normalized first-angular-moment magnitude R and second-moment ratio q_2 from every eligible C40 placement at the same readout, plotted against the normal angle between prescribed initial-peak and cue-centre positions. Opposing local peaks can nearly cancel the first moment: at d20 the two resolved peaks coexist with R=0.0399 and q_2=0.4006. The direction of the activity vector is masked below R=0.1; a small vector is not evidence of absent local activity. D1–D3, Exact native histories selected in the preceding checkpoint: C30,d12 gives a cue-associated single peak; C40,d12 gives one connected region spanning old and cue; C40,d16 retains two distinct associated peaks. Thus matching prescribed physical separation (D1/D2) or prescribed angular separation (D1/D3) does not match finite outcomes when circumference and incumbent architecture also change. D2/D3 provide the controlled within-size placement contrast. Cyan marks cue withdrawal; histories end at 4000 for display while printed readouts use time 14000.

The activity first moment integrates local activity weighted by the outward normal. P is the complex first moment divided by its initial complex value, R=|P|, and q_2 is the absolute second angular moment divided by total active amount. The first moment is an activity summary, without an assumed cell-velocity law or a unique active-site identity. Region association is a spatial endpoint descriptor, not a lineage assignment or proof of asymptotic coexistence. Geometry coordinates and chemical competition are related measurements rather than a causal partition.

Distance-duration response on four supported ring models. A1–A4, Final activity angle after 13,000 cue-free units for selected near/far prescribed cue centres; dotted lines mark the corresponding cue-normal angles. Exact qualified basals, native kinetics and fixed physical support are held within each model. Chemical sizes are C = 40; Spring uses resting C_0 = 2 and actual basal Cb = 2.044. Tested chemical durations are 1/10/100/300/1000, Spring 0.1/1/10/100/1000. At far Holmes T = 1000, R=0.040 masks direction rather than defining a zero angle. B, Native Spring near-cue final profiles show one full-strength region retained at graded translated positions, validated against a rotated exact basal profile. C, Shortest tested passing durations for two explicitly different endpoints: final angle within 15° of the cue normal at R ≥ 0.1, and final original-core excess no more than one half. A dash means no success in the tested duration set. Near

Figure 3. Geometry, cue placement and duration shape retained polarity · continued

Wave-pinning passes the angle gate at 1000 while retaining more than half the original-core excess, illustrating endpoint dependence. These are finite-window observations, not a unique minimum-duration search or migration prediction. Near denotes the selected nearest eligible prescribed placement at the displayed size; it is not a cross-model matched physical distance. Open symbols mark measured failure of the separate full-field final-tail criterion; compact retained chemical T = 1000 references have no invented tail flag. Lines guide the eye between tested points, with intermediate durations unmeasured. Antipodal inputs preserve the initial reflection axis, so 0°/180° far-cue outcomes do not establish a separate binary biological mechanism. The Holmes cue-associated single peak on the first sheet is at C = 30; these duration series use C = 40.

Narrative points and conclusions

- Equal physical separation gives a smaller outward-normal angular difference on a larger ring.
- At fixed Holmes circumference, changing challenger position alters the finite spatial endpoint from one spanning region to distinct old- and cue-associated peaks.
- Strong, opposing local activity can yield a weak global activity vector. The activity angle requires local-pattern and magnitude context.
- Finite near cues can leave one full-strength activity region at a graded intermediate angular position; retained angle need not be a binary old/cue outcome.
- Distance changes regional depletion, retained angle and vector magnitude differently across native models. The shortest tested successful duration depends on the explicitly chosen endpoint.

Proposed measurements and experimental follow-up

Measure physical perimeter and the arc separation of local activity/protrusion regions, and compare their outward-normal angle separately from their local activity strength.

Track local activity profiles alongside first- and second-angular-moment summaries; a weak first moment can accompany multiple strong regions. A cell-velocity law is not assumed.

Measure retained angle, magnitude and local-region activity across characterized cue distances and durations; report angle alignment and regional depletion as separate endpoints.

Figure 3. Geometry, cue placement and duration shape retained polarity · continued

Scope, limitations and pending validation

Physical and angular separation are collinear at fixed circumference. Cross-size comparisons also change incumbent architecture and pool organization, and cannot isolate causal geometry.

Region association is a finite spatial descriptor; no unique site lineage or asymptotic coexistence is established.

The duration grids identify shortest tested passes under explicit criteria; adjacent tested failures and threshold sensitivities remain available, and no optimized minimum duration is claimed. Spring retains its declared mean-force global-extension closure and native stretching, dilution and diffusion.

Model definitions and readouts

$$M_k(t) = \int_0^{C(t)} u(s,t)\exp\left(\frac{2\pi i k s}{C(t)}\right) \mathrm{d}s \tag{3.1}$$

The native active-field angular moments use physical arc quadrature. M_0 is the instantaneous active amount; C(t) is the actual circumference, including Spring expansion and contraction.

$$P(t) = \frac{M_1(t)}{M_1(0)}, \qquad R(t) = |P(t)|, \qquad q_k(t) = \frac{|M_k(t)|}{M_0(t)} \tag{3.2}$$

P measures the first moment relative to its initial complex value. The shape moments q_1 and q_2 use instantaneous total active amount; they are distinct from R. Direction requires the declared magnitude gate.

$$\Delta\theta_0 = \frac{2\pi d_0}{C_b} \tag{3.3}$$

Prescribed initial-peak-to-cue separation d_0 is mapped to the normal-angle difference using the basal circumference Cb. This cue-placement quantity differs from subsequent activity-site relocation.

Sources and reproducibility

Geometry evidence and negative results

Complete placement outcomes and moments

Held-out-size descriptor transfer

Historical chemical-only gain/recovery and interval/ring sheet

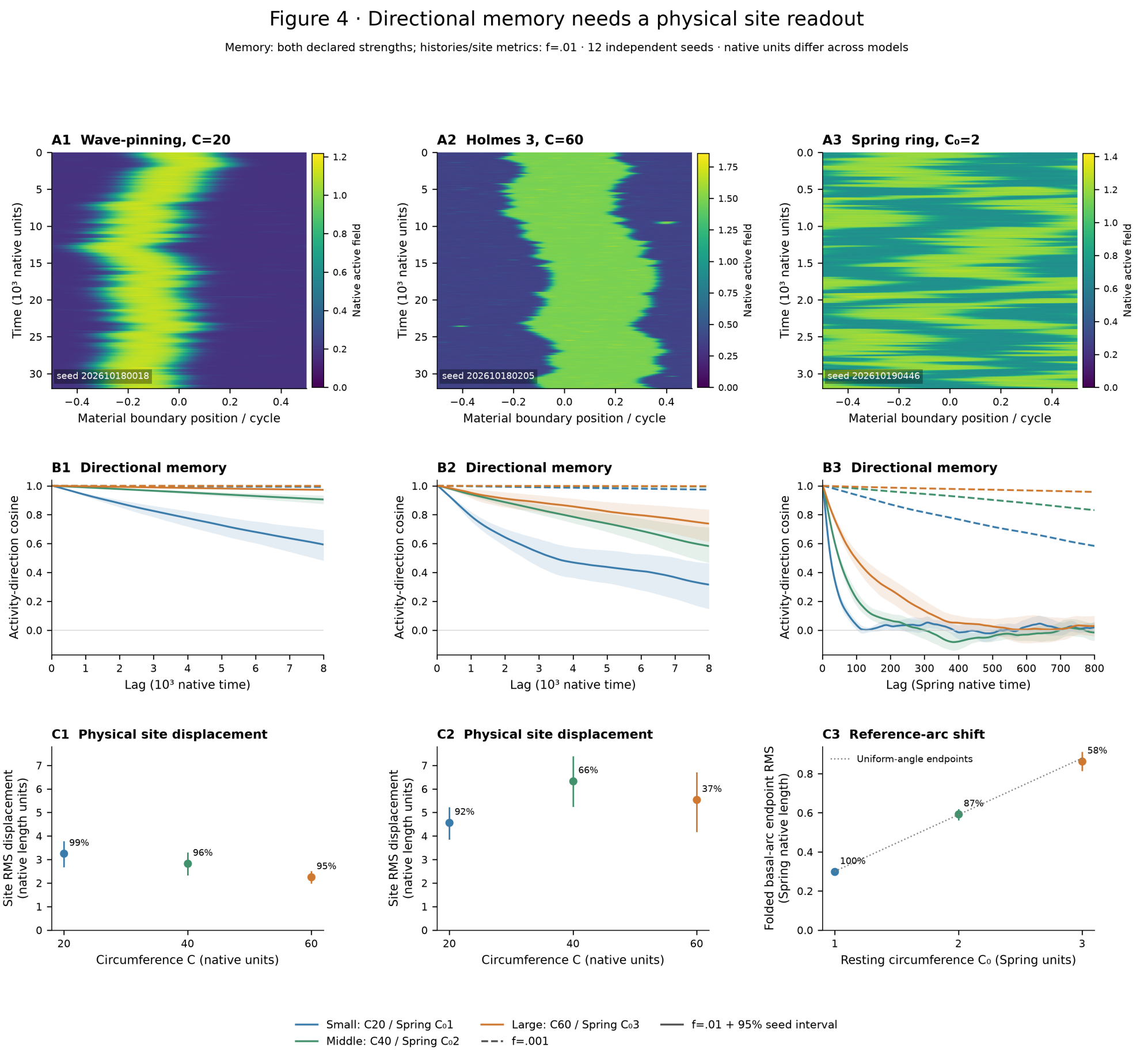
Numerical and provenance validation

Four-model finite-duration continuation and endpoint limits

Figure 3. Geometry, cue placement and duration shape retained polarity · continued

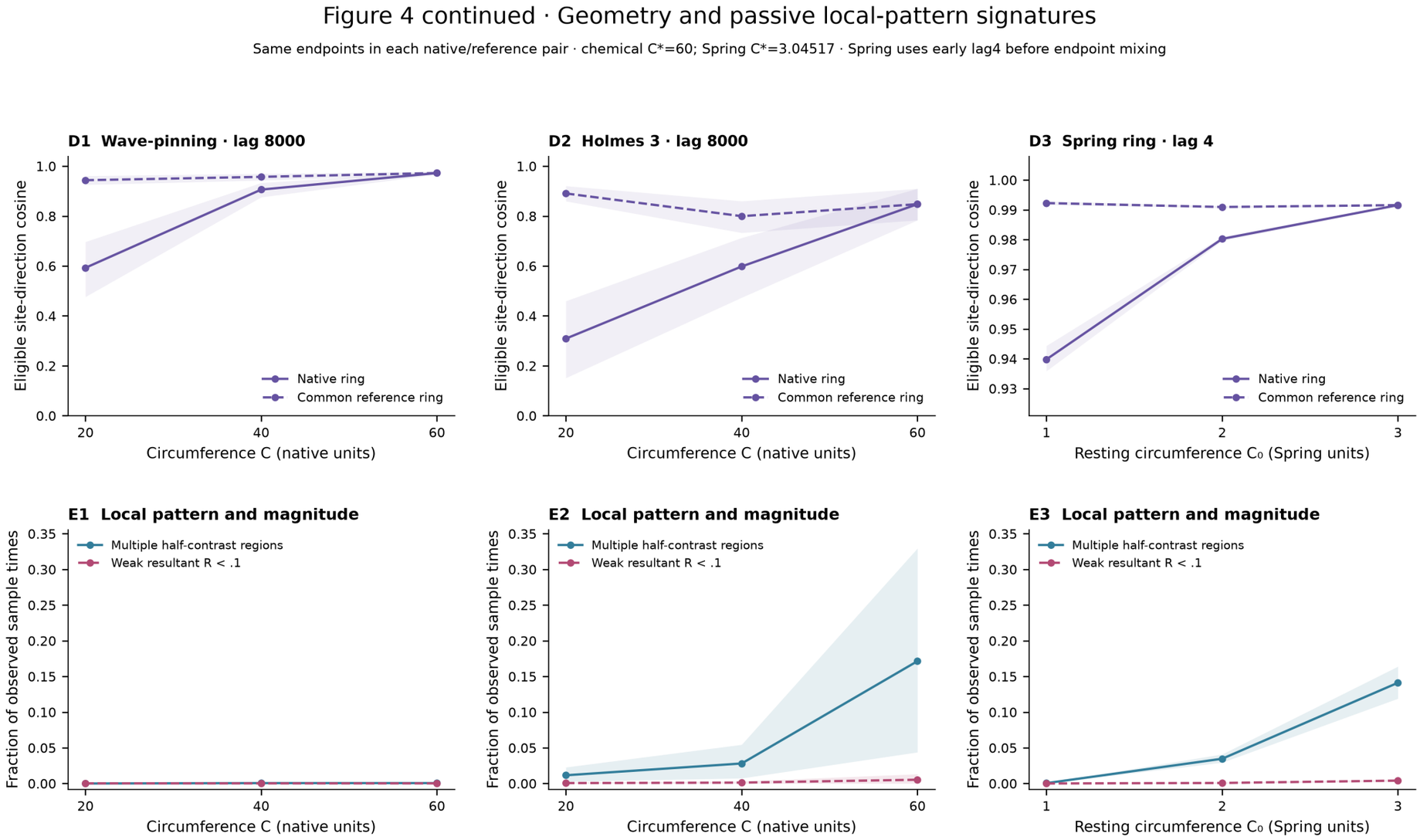
Complete duration grid with actual native archive paths
Independent chemical/Spring retained-profile checks
Spring descriptor coverage and nonuniform interval-correspondence scope
Four-model duration evidence and limits
All150requested duration conditions
Endpoint sensitivity and all tested success/failure sets
Spring native field and geometry audit
Exact shared Spring preparations and qualification
Independent Spring preparation windows and holdout
Spring basal function verification
Spring challenged-state function verification
Declared Spring ring model and approved geometric carryover
Spring categorical invariance, continuous descriptors and interval-correspondence scope
Spring basal organization, input gain and local recovery
Spring common active-loss response at exact challenged t10 states
Declared Spring ring model and numerical validation

Figure 4 · Directional memory and physical site dynamics · image 1 of 2



Native histories, activity-direction memory and physical/reference-arc site shifts.

Figure 4 · Directional memory and physical site dynamics · image 2 of 2



Paired geometric remapping and passive local-pattern/magnitude readouts.

Figure 4 · Directional memory and physical site dynamics

stochastic dynamics

Detailed legend

Figure 4. Directional memory needs a physical site readout. Memory panels B show both predeclared strengths: dashed f=.001, solid f=.01. Histories and site/pattern metrics use f=.01. Every noisy group retains all 12 independent seeds; shading and bars are pointwise 95% whole-seed intervals (B shades the stronger input). Native units differ across models.

A1–A3: native active-field histories from the predeclared condition medoids: Wave-pinning C20, Holmes C60 and Spring $C_0$2. Each model uses a zero-based native activity scale; boundary position is material position around the ring.

B1–B3: whole-window activity-direction cosine, conditional on both endpoint relative first-moment magnitudes R≥.1. The physical activity moment $M_1(t)$ is divided by its initial complex value. Activity direction is distinct from cell movement.

C1–C3: dominant half-contrast-site endpoint RMS shift at lag 8000 for chemical rings or 800 for Spring; percentages are eligible endpoint pairs. Spring uses basal-reference arc shift Cbasal×wrapped material-position change, with the dotted uniform-angle reference Cbasal/√12. Long-lag folded shifts can approach circular mixing and are not travel distances. Spring has its own native length axis.

D1–D3: native site-angle cosine and remapping of the same eligible shortest shifts to common circumference C*=60 for chemical rings or 3.04517 for Spring. Chemical panels use lag 8000; Spring uses early lag 4 before late endpoint mixing. Full lag curves and 4/50/800 comparisons remain supporting evidence. This is a geometric coordinate comparison, not a causal decomposition of global activity-vector memory.

E1–E3: observed time fractions with multiple current half-contrast activity regions and weak relative resultant R<.1. Region multiplicity does not establish functional competition. All 6401 scalar samples per seed enter each fraction. Supporting increments retain rare >π/4 Spring changes at cadence .5 and none in chemical traces at cadence 5, conditional on both endpoint R≥.1; these sampled diagnostics do not define an event class or exclude faster changes.

Complete Goryachev results remain supporting evidence: its high-input C20 mean control also loses polarity, precluding an intact-state wandering interpretation. Both declared strengths, all mean/no-input controls and medoids, magnitude and increment distributions, eligibility sensitivities and full/half-window analyses are retained. Methods define Spring's approved stretch/dilution/diffusion carryover, mean-force global-extension closure and finite transported-material noise law.

Narrative points and conclusions

Figure 4 · Directional memory and physical site dynamics · continued

- Angular-memory and physical site-shift measurements can show different size trends. Spring makes the forcing dependence explicit: weak-input lag-800 direction cosine rises .584/.832/.958 across sizes despite nonmonotone basal-arc RMS, whereas stronger input gives .019/−.017/.028 with substantial mean activity remaining and late folded shifts near the uniform-angle reference. All three main models retain their native units and controls.
- Pairwise coordinate remapping exposes the geometric part of site-angle retention, while local-region and magnitude readouts distinguish observable organizations without classifying abrupt events or inferring competition.

Proposed measurements and experimental follow-up

Activity-direction correlation and full-vector memory

Physical/reference-arc site RMS with endpoint coverage

Native/common-reference site-angle retention on identical pairs

Local-region and weak-resultant occupancy

Gated increment distributions and full/half-window sensitivity

Scope, limitations and pending validation

Native units are not calibrated across models.

Activity direction does not assign movement.

Endpoint pairs do not establish site ancestry.

Spring reference-arc shifts separate material change from uniform stretch.

Remapping is a coordinate comparison, not a causal decomposition.

Finite-knot/noise-law and finite-window limitations remain explicit.

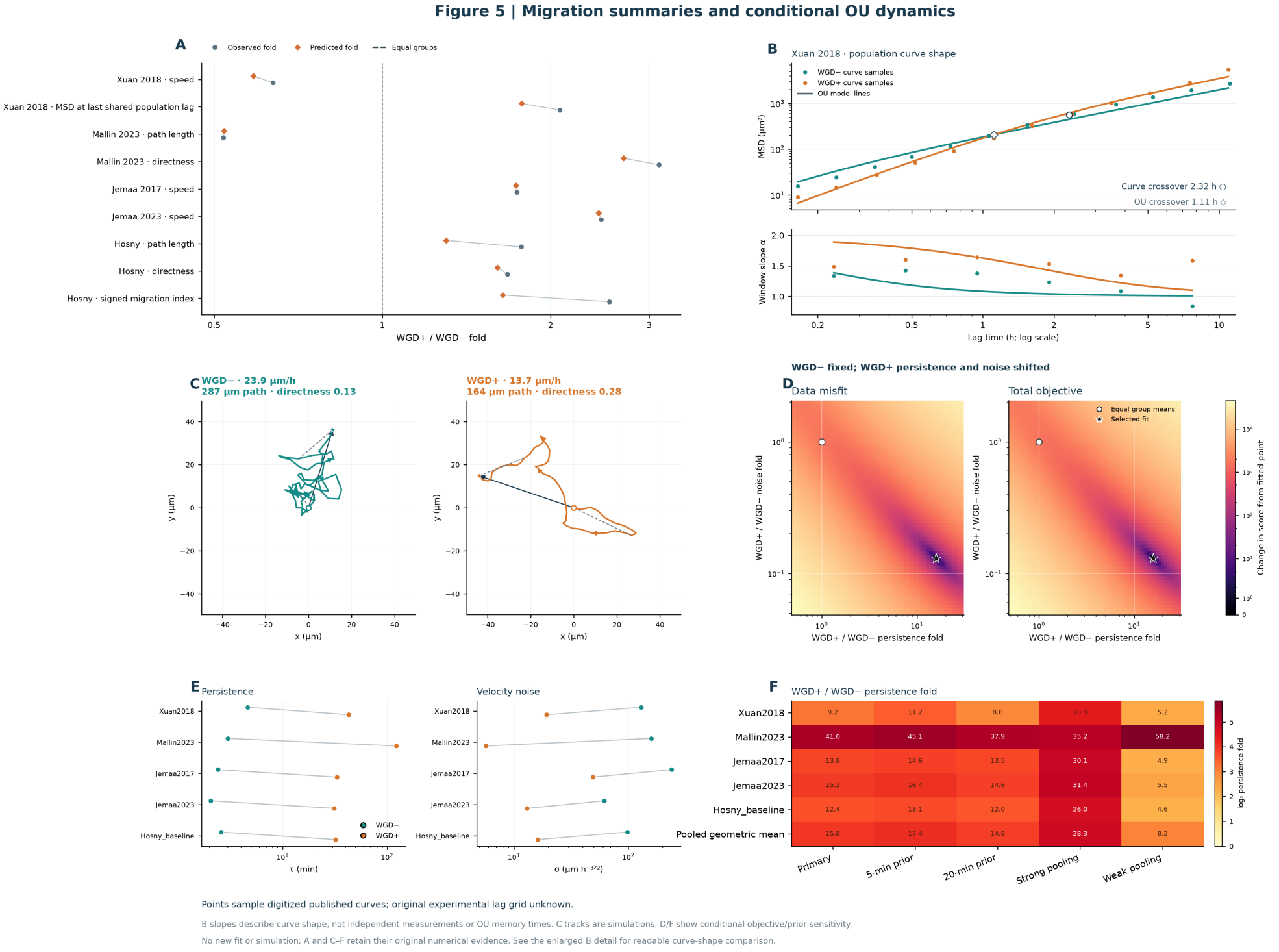
Sources and reproducibility

Original chemical-ring stochastic report
Original noise design
Complete Goryachev/chemical interim figure
Frozen Spring main design
Paired finite-law estimator sensitivity
Spring full analysis audit
Spring centroid/half-window/CDF/forcing audit
Spring full seed coverage
Spring methods and limits

Figure 4 · Directional memory and physical site dynamics · continued

Final four-model quantitative evidence
Spring raw full/half-window memory
Spring site and stretch metrics
Spring first/last-half site metrics
Spring magnitude-gated increment distributions
Spring actual forcing audits
Spring early/late endpoint summary

Figure 5. Migration observations and conditional OU interpretation · image 1 of 2



Reassembled Figure 5: original A and C–F numerical evidence, revised MSD/curve-shape panel B.

Figure 5. Migration observations and conditional OU interpretation · image 2 of 2

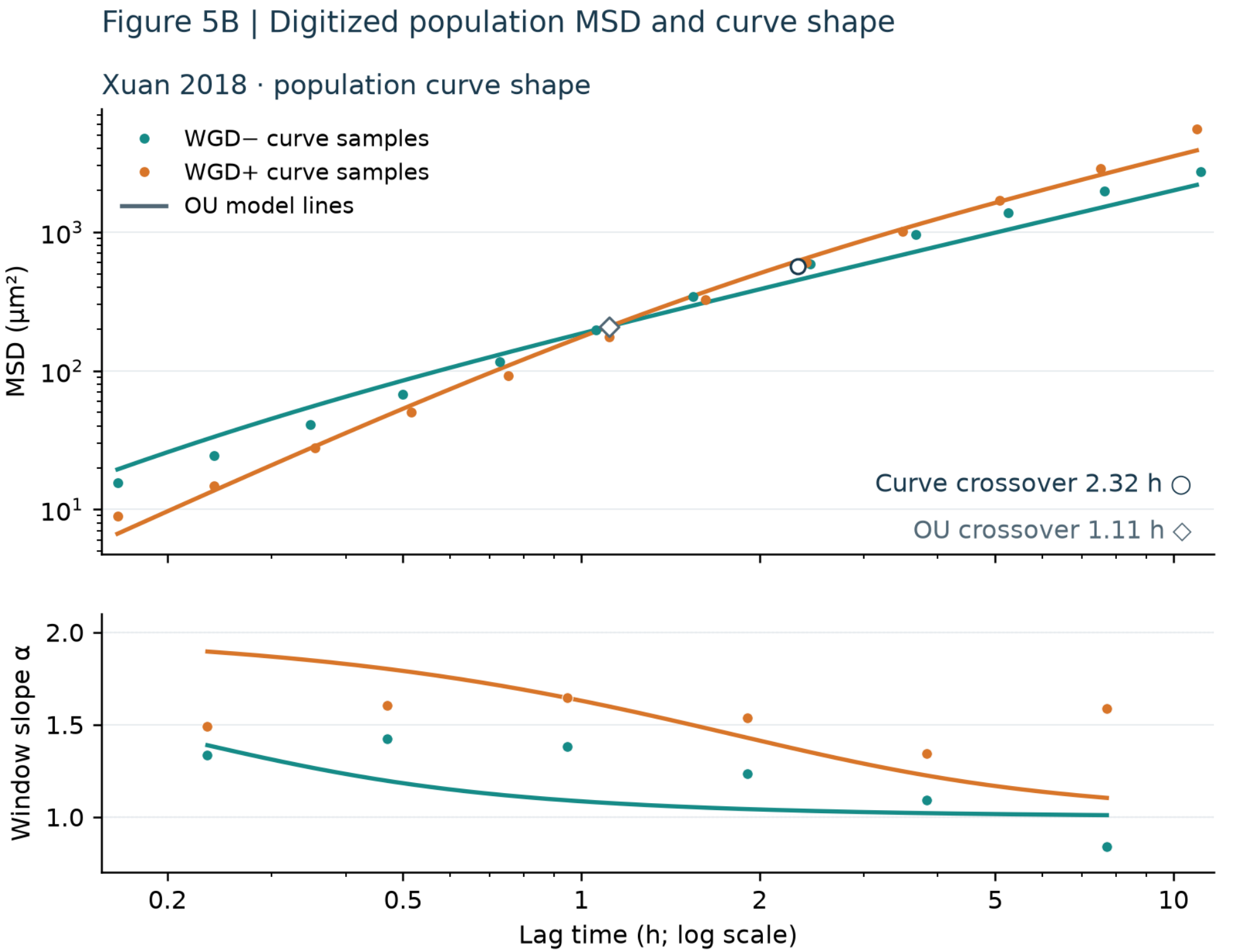


Figure 5B continued: enlarged detail with explicit raster-sample caveat.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Existing empirical support, with observed migration summaries kept separate from fitted velocity-memory parameters.

Detailed legend

A, Published/digitized enlarged or polyploid-group to reference-group ratios for nine migration summaries, alongside primary Ornstein–Uhlenbeck (OU) fit predictions. Xuan contributes sampled mean speed and population mean-squared displacement (MSD) at the last shared lag, approximately 10.95 h; Hosny summaries are group medians. Mallin central-bar statistic is not specified by the source and retains the historical mean approximation. These distinct source summaries are not interchangeable measures of persistence.

B, Upper: twelve chosen display samples from each digitized Xuan population MSD curve (small points) and the unchanged fitted analytical OU expectation (solid lines), on logarithmic lag and MSD axes. These are samples of the published raster curves, not recovered experimental lag positions or independent cell observations; the original experimental lag grid is unknown. The open circle identifies the crossover of the full digitized traces at approximately 2.32 h, computed by linear interpolation in native lag and MSD. The open diamond identifies the different fitted OU crossover at approximately 1.11 h. Lower: descriptive slopes of log(MSD) versus log(lag) fitted in six equal log-lag windows of the full digitized curves (points), and matching-window moving OU slopes (solid lines). Each window spans approximately 0.304 decades; both observed and model slopes use the same full window support. Observed raster columns remain correlated and give no biological sample size or confidence interval. These slopes are newly calculated curve-shape summaries, not author-reported local exponents or fitted OU memory times. A constant power-law exponent would appear horizontal; reference values 1 and 2 indicate diffusive and straight-line scaling. The model does not reproduce the complete observed curvature or the observed crossover. The analytical fit, all fitting targets and numerical inputs to A and C–F are unchanged.

C, Representative Xuan primary-fit model-generated tracks, with sampled speed, total path length and directness (net displacement divided by path length). Both groups use identical spatial scales. The tracks are simulations, not published individual-cell trajectories; a slower sampled trajectory can nevertheless have greater endpoint directness. Dashed chords illustrate the contribution of finite-time displacements.

D, With reference-group parameters fixed, vary enlarged-group persistence and velocity-noise ratios. Separate surfaces show change in the descriptive data-misfit score and in the total regularized objective; the selected fit and equal-group reference are marked. The heuristic score is not a calibrated likelihood, and the surfaces are not confidence regions.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

E, Primary paired persistence and velocity-noise estimates for Xuan 2018, Mallin 2023, Jemaà 2017, Jemaà 2023 and the Hosny hypoxic baseline. F, Enlarged/reference persistence folds under the primary 10-minute prior, 5- and 20-minute alternatives and stronger/weaker pooling, with the pooled geometric mean. Priors and pooling define conditional sensitivity scenarios, not uncertainty intervals. The Jemaà comparisons constrain speed alone, leaving persistence/noise tradeoffs dependent on hierarchy and priors.

The fitted population contrasts span stable tetraploids, giant polyploids and treatment-induced polyaneuploid cancer cells (PACCs). The inherited WGD+/WGD− labels in the artwork are comparison shorthand and do not establish matched WGD-only interventions in every study. Hosny observations derive from the recovered preprint; equivalence to the journal version remains unresolved.

Narrative points and conclusions

- Xuan combines lower sampled speed (0.64-fold) with greater late-lag population MSD (2.08-fold); Mallin combines shorter path length (0.52-fold) with greater directness (3.12-fold). Persistence-related behaviour therefore deserves analysis separately from speed.
- Hosny baseline PACCs have greater median path length (1.77-fold) and directness (1.67-fold). Signed migration index is a separate directional summary.
- All five primary OU comparisons infer longer velocity memory. The pooled primary persistence fold is about 15.8, and remains above one across the declared prior/pooling scenarios; this is a conditional model result.
- The fit does not reproduce the complete Xuan curve shape within its raster envelope, and data-only sensitivity has rank 18 of 22. Optimization completion does not establish model adequacy or uniquely identified persistence times.
- The observed Xuan curves cross near 2.32 h while the unchanged fitted OU curves cross near 1.11 h. Matched-window slopes make the remaining curve-shape discrepancy visible; fitted persistence is a conditional summary.
- Lower sampled speed can coexist with greater late-lag displacement in the enlarged-group comparison. Its causal connection to polarity mechanisms requires matched activity and migration measurements.

Proposed measurements and experimental follow-up

The new WGD assay should measure matched trajectories directly, with declared sampling and observation duration, and assess speed, displacement, directness and lag-dependent MSD before considering fitted persistence. An activity-polarity memory

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

measurement would complement these quantities without equating activity and velocity.

Scope, limitations and pending validation

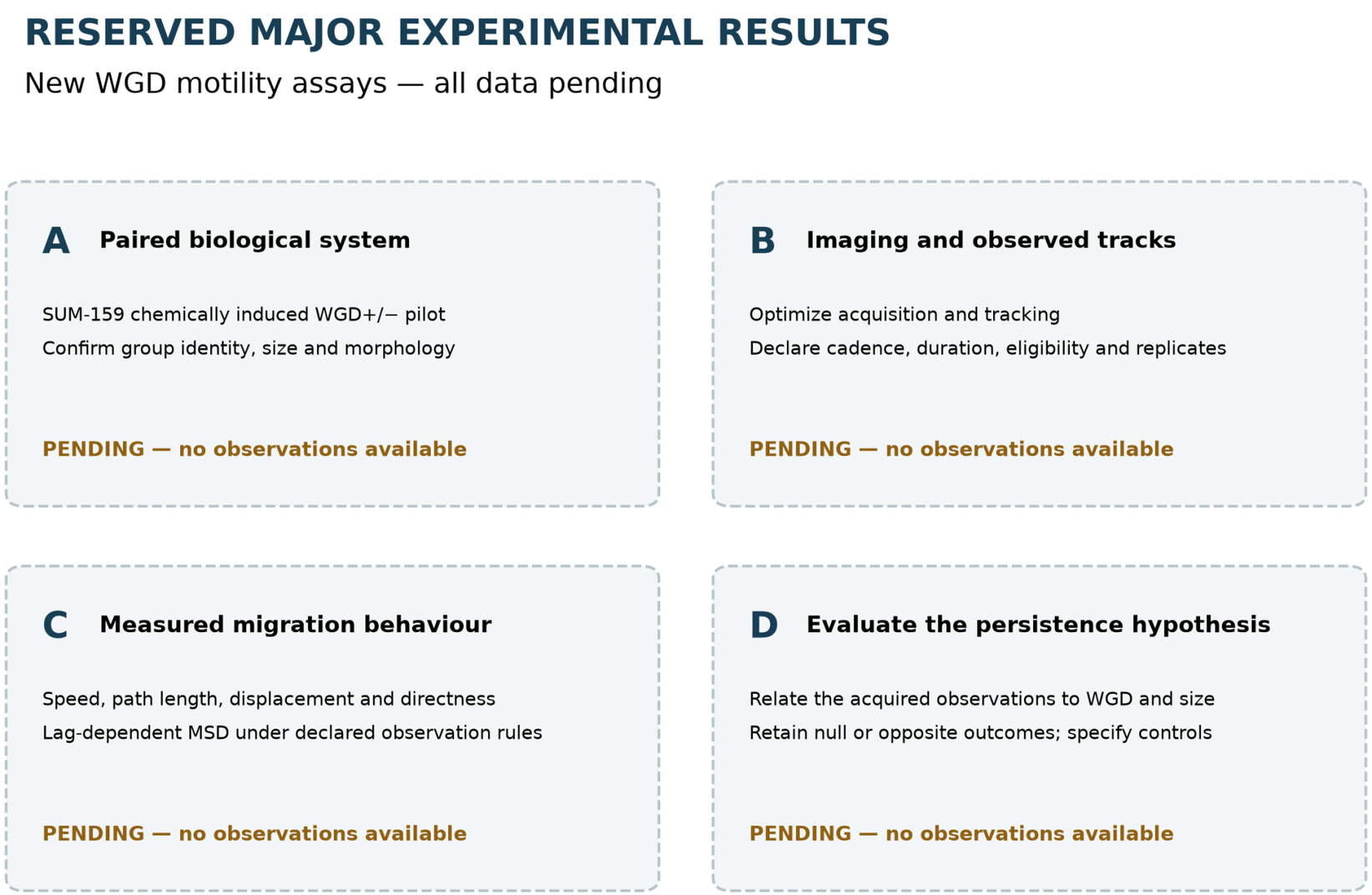
Neither these heterogeneous group comparisons nor the OU fits establish that enlargement alone causes persistence. Cell identity, induction/treatment and acquisition differences require explicit treatment in the new experiments.

The requested observed-point/model-line and curve-shape display is implemented. The original experimental lag grid remains unknown; neither display samples nor raster-derived slopes supply independent biological replicates. The observed/model crossover discrepancy remains visible.

Sources and reproducibility

Refined OU analysis
Original panel legend
Observed folds
Fit results and adequacy
Hosny sensitivity supplement
Revised display, exact preserved inputs and direct-user feedback
Migration display numerical and preservation checks

Figure 6. RESERVED: new WGD motility results



Explicit placeholder. Every panel is pending acquisition, analysis and author review; no mock observations are displayed.

Figure 6. RESERVED: new WGD motility-assay results

A substantial new experimental results section and main figure. No observations are available; the panel layout is a proposal, not an experimental result.

Detailed legend

A, Proposed experimental context: the SUM-159 chemically induced WGD+/− pair is the agreed starting pilot with the Speros collaboration. Imaging should first be optimized for that pair before expansion to additional cell lines. Ploidy/group identity, physical size and morphology need to be documented so the intended comparison is clear.

B, Reserved for representative experimental fields and trajectories under the selected acquisition conditions. The eventual panel should state sampling interval, usable duration, spatial scale, track eligibility and exclusions. Number of independent preparations and biological replicates remains to be specified.

C, Reserved for empirical motility summaries: sampled speed, accumulated path length, endpoint displacement/directness and lag-dependent MSD. The final readouts, follow-up window and analysis plan are to be settled before interpreting the data; this placeholder does not imply that all listed measurements will be technically feasible.

D, Reserved for the measured relationship between WGD-associated enlargement and migration behaviour, including size distributions and controls needed to interpret the group contrast. Any activity/protrusion measurements arising from the mechanism proposals require a separately agreed assay and should be distinguished from the basic motility pilot. Statistical models, uncertainty reporting and sample-size targets remain open.

Narrative points and conclusions

- This major results section will test whether the planned paired experimental system exhibits the migration behaviour motivating the theoretical work.
- Its conclusions must follow the acquired observations, including null or opposing results. Simulation output and literature fits cannot fill this gap.
- Agreement between increased polarity resistance or activity memory and migration persistence would support the proposed connection; causal attribution requires appropriate perturbations and controls.

Proposed measurements and experimental follow-up

Pending decisions include plating density, acquisition duration and cadence, field/condition throughput, tracking and exclusion rules, replicate structure, confirmation of WGD status, size quantification and the statistical analysis. The known throughput versus spatial/temporal resolution trade-off should be resolved by the pilot.

Figure 6. RESERVED: new WGD motility-assay results · continued

Mechanism validation can subsequently examine incumbent extent, early challenger growth, local loss/rebound, site fates and global activity-polarity correlation under characterized perturbations. These activity experiments are proposals alongside the theoretical mechanisms and are separate from the reserved motility-results contribution.

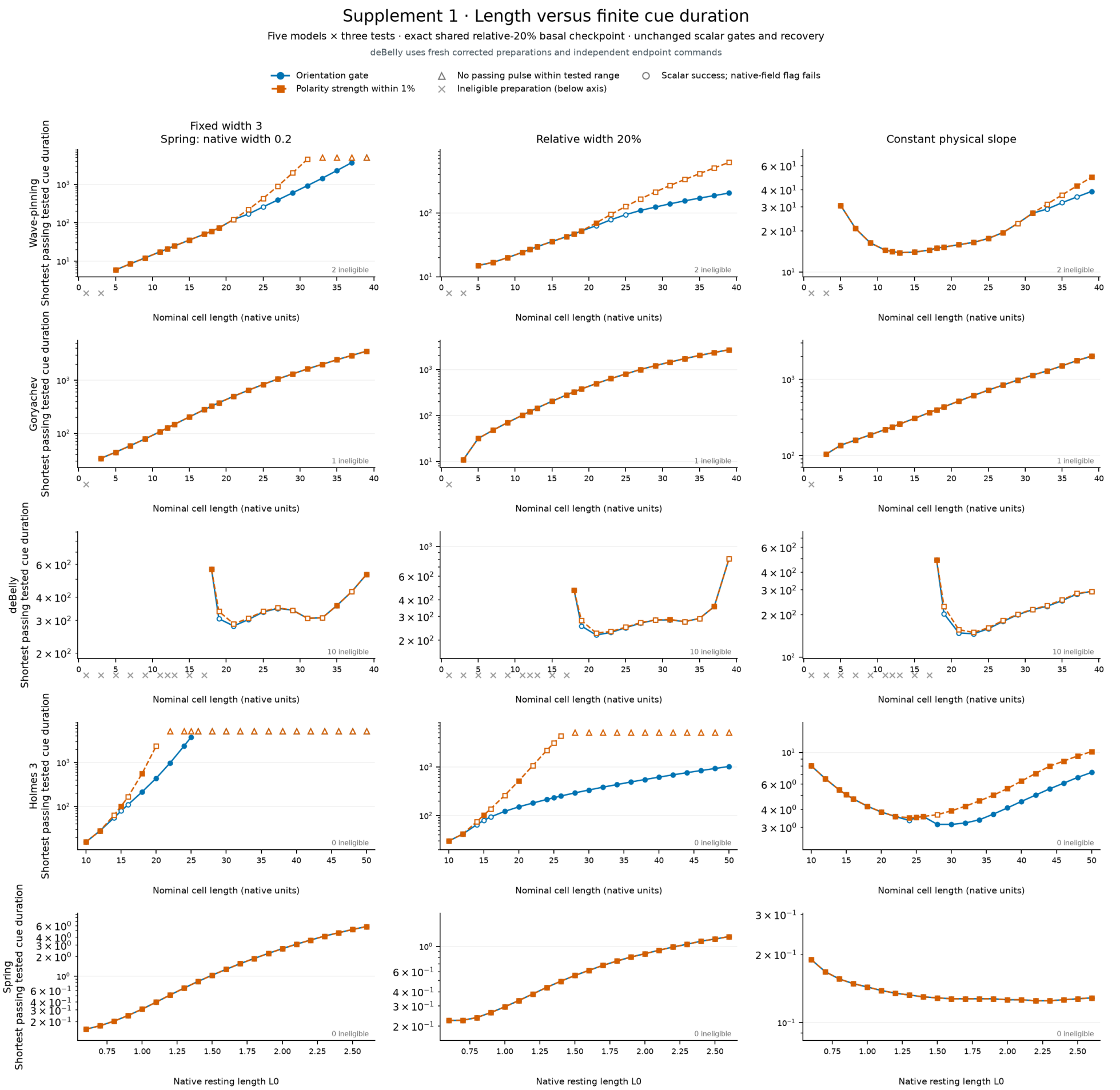
Scope, limitations and pending validation

No pilot acquisition, sample size, effect estimate, significance claim or experimental outcome is supplied by the available evidence.

Sources and reproducibility

Author manuscript vision and supplied collaboration context

S1. Full cue-duration response curves



Both patches use a constant peak across all lengths; Goryachev K=64, peak=6.4. Gradients retain one constant physical slope per model.
Each positive duration restarts the original checkpoint. Full strength adds the 1% reversed initial-polarity band to the orientation gate.
Open markers retain failing field-tail flags. Censoring marks explicit tested limits; missing endpoints break lines. Native times and scales differ.
Data generated 2026-10-08T19:26:33.200718+00:00; producer finished. Every imported and fresh trial passed independent native-array audit.

Canonical corrected five-model duration curves; source working number Supplement 1.

S1. Full cue-duration response curves

Essential curve context for the main comparison heatmap.

Detailed legend

Five model rows and three finite-cue columns retain the complete sampled length dependence. Coloured curves distinguish orientation from strength replacement. Open success markers retain scalar successes that fail the separate native-field tail flag; triangles indicate no passing pulse within the tested range; crosses indicate ineligible preparations. Missing values are not zero response times. Each duration excludes preparation and the subsequent 10,000-unit basal recovery. Spring uses native length and time. See Figure 1 for exact gates and stimulus conventions.

Narrative points and conclusions

- Corrected deBelly is nonmonotone across all three geometries. The positive rank coefficients summarize association rather than its full curve shape.
- Constant-slope Spring shows the opposite association to its patch responses. Holmes orientation under that cue varies with little monotone association. Censoring and native-field flags restrict interpretation even where rank correlation is strong.

Sources and reproducibility

Current figure legend and gate definitions

Supplementary material and supporting analyses

S2. Numerical qualification and preparation provenance

Exact common starting states, duration-boundary audits, corrected deBelly endpoint controls and selected mesh diagnostics; source histories remain distinguishable from compact committed summaries.

polarity1d/manuscript/THREE_TESTS.md

S3. Mechanism and stochastic sensitivity

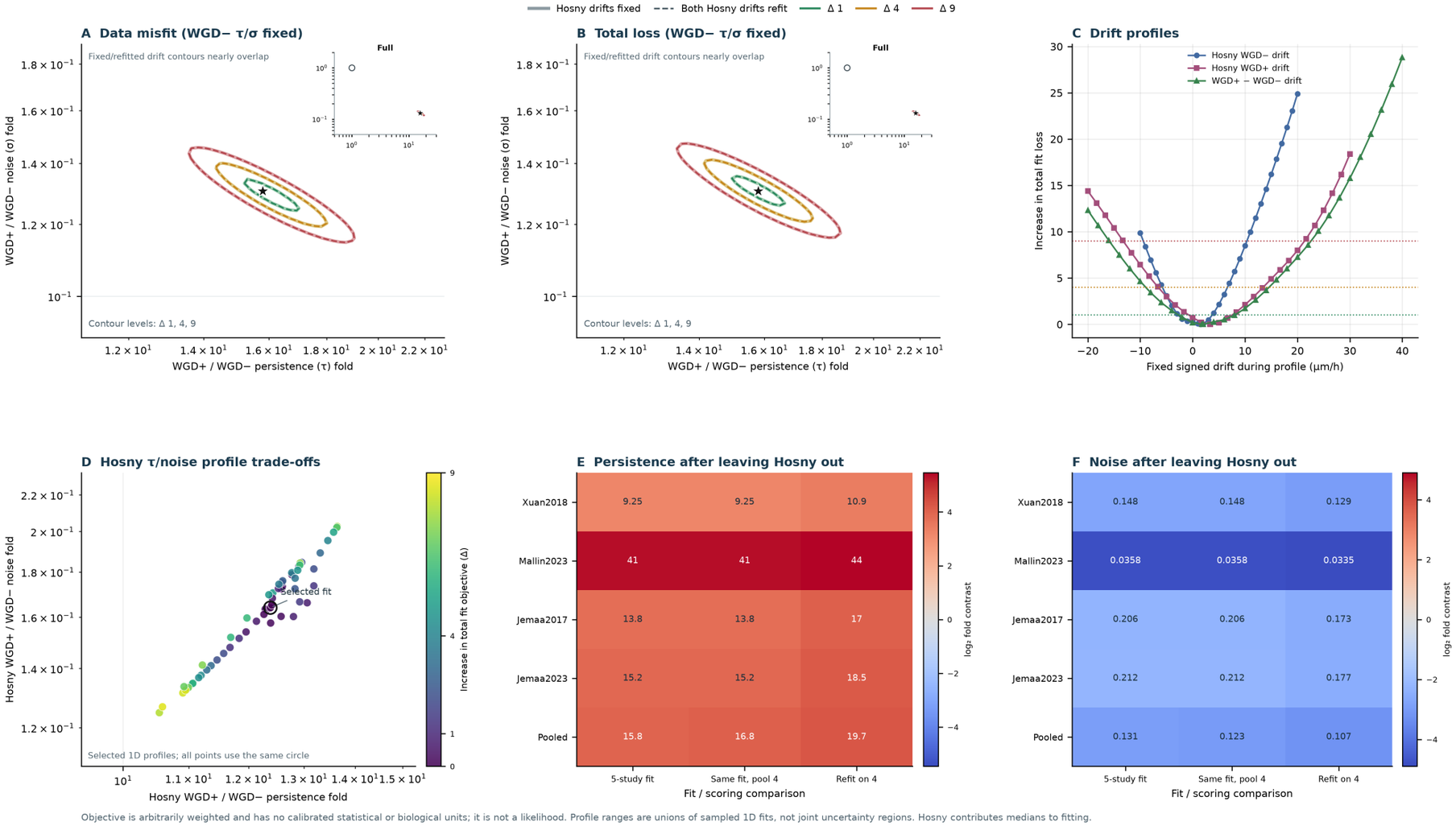
Complete gain/recovery grids, clock qualification, finite-cue definitions, outcome tables, site/vector diagnostics and numerical/noise controls accompany the mechanism checkpoint. Revised Figure 4 also displays both strengths, complete half-window diagnostics, physical-site eligibility and matched C20/C60 native examples; compact source tables support portable rerendering.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/VISION_CHECKPOINT_REQUIREMENTS.md

S4. Migration-model adequacy and Hosny sensitivity

How much does the Hosny drift assumption change the fitted contrast?

Primary Gaussian OU hierarchy · profile panels reoptimize remaining parameters · Δ values are descriptive fit sensitivity



Objective is arbitrarily weighted and has no calibrated statistical or biological units; it is not a likelihood. Profile ranges are unions of sampled 1D fits, not joint uncertainty regions. Hosny contributes medians to fitting.

Existing Hosny sensitivity supplement from the refined OU analysis.

S4. Migration-model adequacy and Hosny sensitivity

Supporting conditional sensitivity analysis; it is not an independent biological replication.

Detailed legend

The retained supplement explores signed-drift and leave-Hosny-out sensitivity within the declared OU hierarchy. Its fitted changes and descriptive-score profiles are conditional on the model and supplied summaries, not confidence intervals. Data-only identifiability, Xuan population-curve residuals and prior/pooling assumptions remain part of the adequacy assessment. Representative movie diagnostics are separate: acquisition timing is uncertain and nuclear appearance does not establish ploidy.

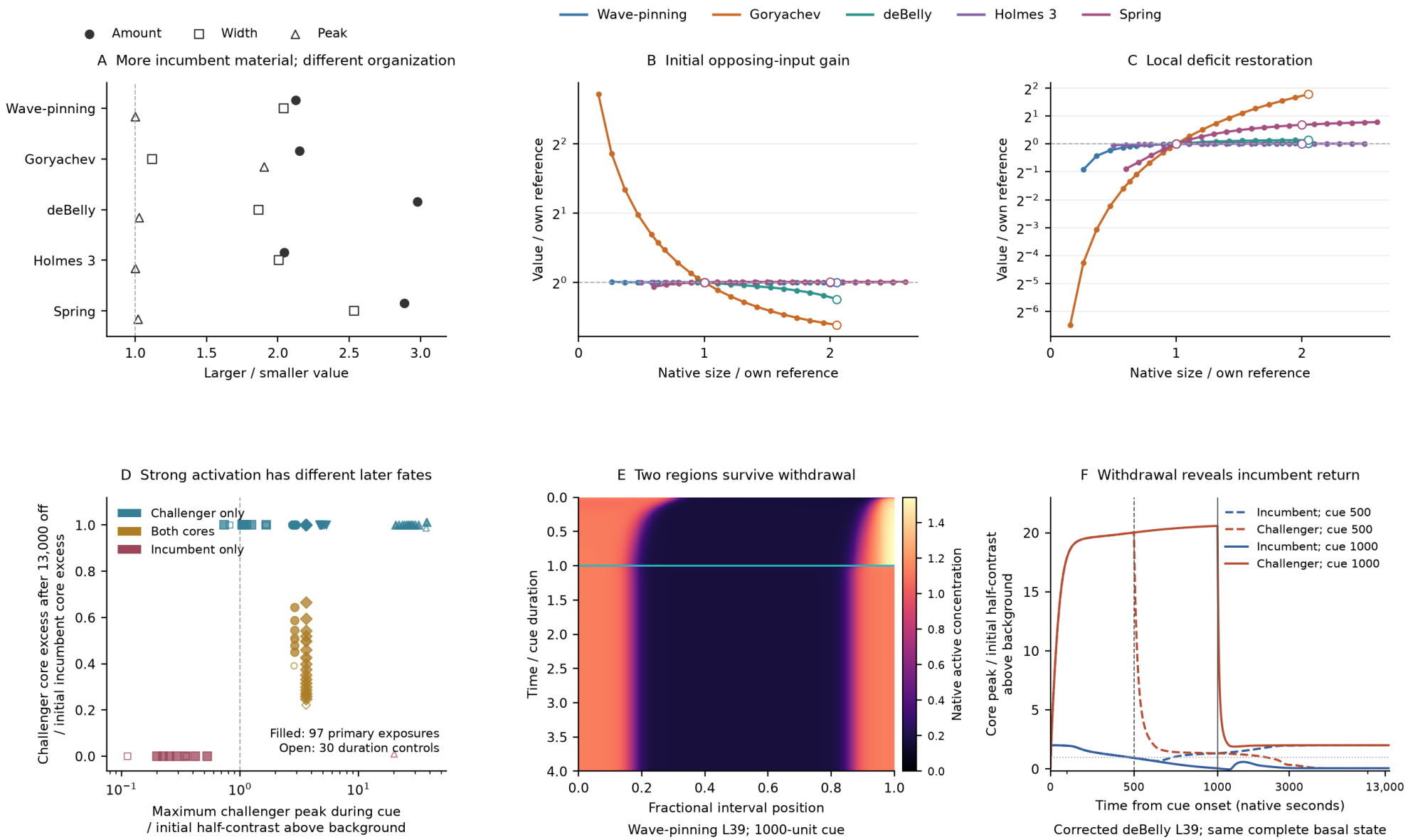
Sources and reproducibility

Original Hosny supplement caption

Sensitivity methods and results

All-five response context and post-withdrawal ambiguity

Figure 2 supporting evidence | All-five organization, response and post-withdrawal ambiguity



A: near-doubling pairs L19→39; Holmes 20→40; Spring $L_0$1→2. B–C: all 97 qualified basals; native units and clocks differ.
D: core geography is threshold-defined, not a scalar replacement gate or tracked site identity. E: cyan marks withdrawal.
F: dashed 500-unit cue returns to the incumbent; solid 1000-unit cue retains the challenger. Each has 13,000 cue-free units.

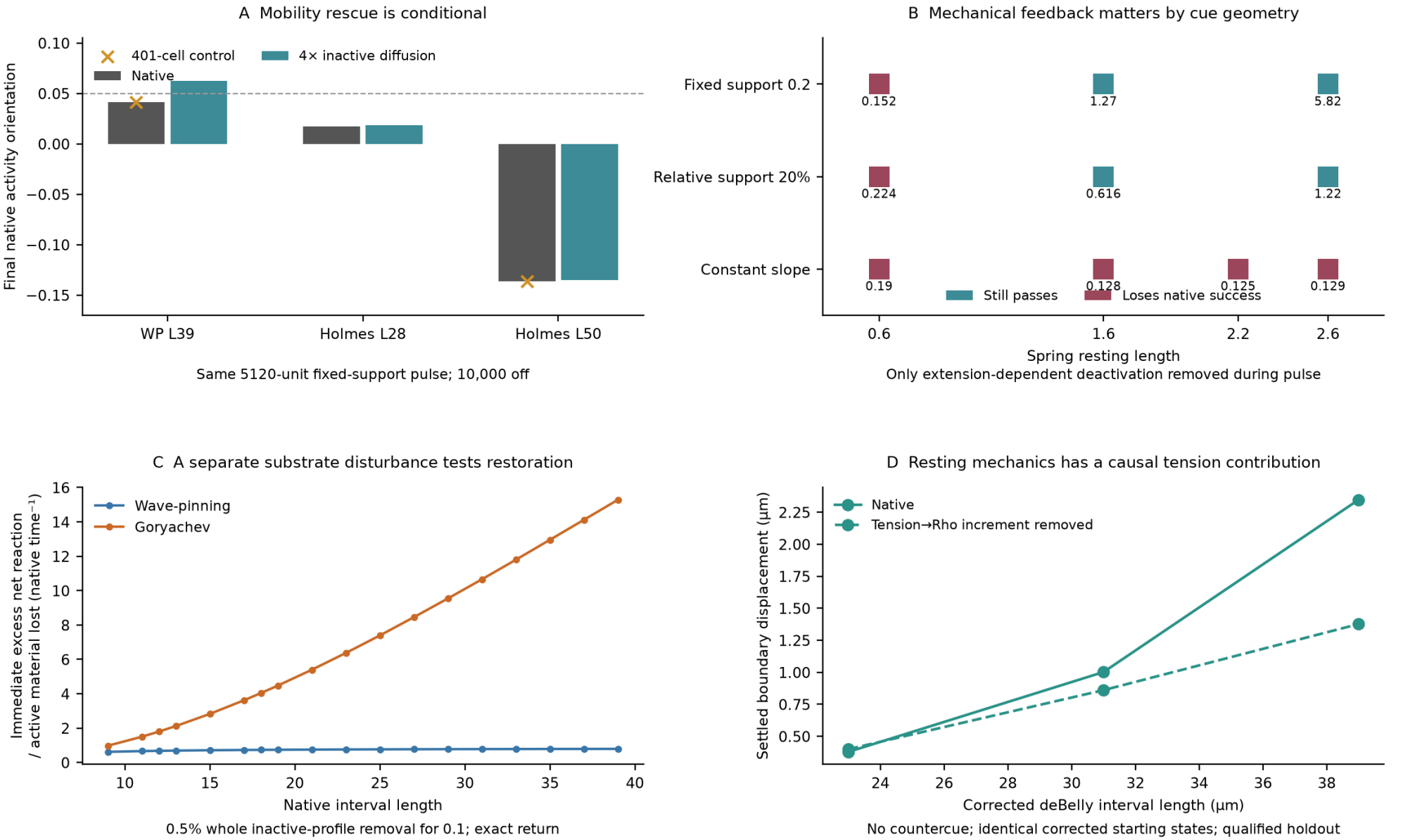
All-five response context and post-withdrawal ambiguity

Detailed legend

All-five organization/gain/recovery, all 127 native cue/fate observations and exact corrected deBelly39 duration contrast: strong challenger activation followed by incumbent return after 500 units, versus challenger retention after 1000, with 13,000 cue-free units in both. Symlog time is linear through 1000 then logarithmic.

Selected evolved interventions

Figure 2 supporting evidence | Selected interventions distinguish specific contributions



A–B are fixed-dose evolved controls, not new intervention-specific duration thresholds. B labels give each native successful pulse duration; native feedback returns for recovery. C is a different perturbation from the common active-loss assay. D concerns resting organization. These interventions do not assign additive causal fractions to polarity replacement or establish a shared physical clock between models.

Selected evolved interventions

Detailed legend

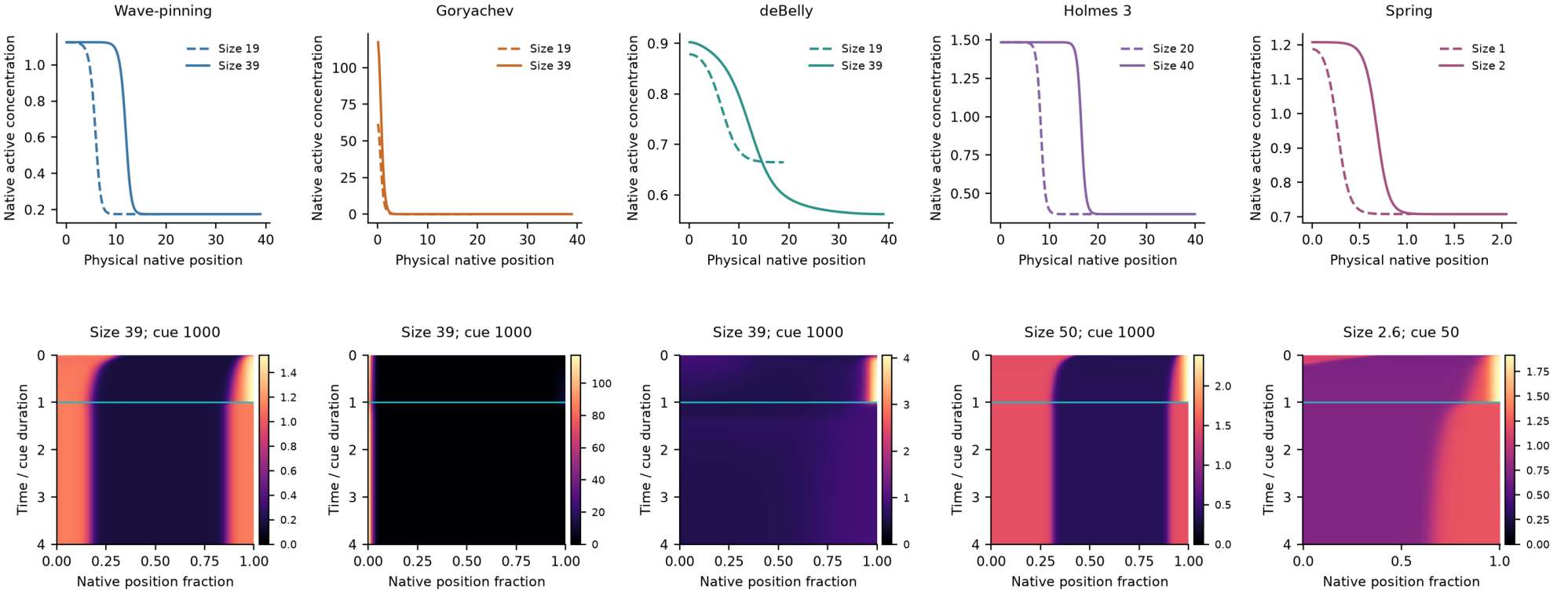
WP L39 retained-orientation control: fourfold inactive diffusion throughout cue and recovery changes final orientation from 0.04146 (fail) to 0.06215 (pass) after the same 5120-unit pulse. Both conditions fail main panel A's stronger full-polarity-strength gate. The sheet also retains cue-specific Spring feedback removal, separate Goryachev/WP inactive-bolus restoration and corrected deBelly resting ablation. Exact protocols and limits are in the linked evidence file.

Scope, limitations and pending validation

The WP orientation-gate result does not establish a change in panel A's strength threshold. Fixed-dose controls do not locate new duration thresholds; resting ablation is separate from reversal causation.

Native profiles and illustrative histories

Figure 2 supporting evidence | Native basal organization and selected complete-state histories



Top: exact near-doubling basal profiles, physical native coordinates. Bottom: prespecified largest qualified size in every model; zero-based, model-specific native activity scales; cyan = withdrawal. Spring position is material fraction. Full post-cue horizon is 13,000. The fixed-support corrected deBelly history is distinct from the prescribed relative-20% mechanism illustration in the retained source report.

Native profiles and illustrative histories

Detailed legend

Exact basal profiles at the stated near-doubling pairs and prespecified largest qualified size in each model.

Supplementary material and supporting analyses

Historical chemical-only placement grid; Spring counterpart linked separately

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/revision_figure3_all_outcomes.png

Historical chemical-only descriptor transfer; no invariant-class Spring score

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/revision_figure3_descriptor_transfer.png

Current Spring long-cue invariant class and continuous descriptor coverage

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/spring_distance/supporting/data/spring_standard_long_cue_descriptor_coverage.csv

Complete fixed-duration distance response in all 652 native chemical rings

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/distance_response/distance_response_all_sizes.png

All three Spring sizes and five finite-duration placements

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/spring_distance/spring_distance_duration.png

All 150 requested duration conditions and actual native archive provenance

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/distance_response/supporting/data/distance_duration_all_models.csv

Chemical 75-condition duration grid including exact retained references

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/distance_response/supporting/data/chemical_distance_duration_with_reference.csv

Endpoint threshold sensitivity and complete tested success/failure sets

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/distance_response/supporting/data/distance_duration_shortest_tested_sensitivity.csv

All original-core and activity-alignment crossing brackets

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/distance_response/supporting/data/distance_crossing_brackets.csv

Supplementary material and supporting analyses · continued

Spring transported-background core and alignment crossing brackets

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/spring_distance/supporting/data/spring_distance_crossing_brackets.csv

Spring basal organization, gain and common active-loss recovery

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/spring_function/README.md

Spring challenged-state conditional regional function measurements

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/spring_challenged_function/README.md

Interactive distance-response explorer

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/supporting/viewers/distance_response.html

Spring: every native medoid

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_all_histories.png

Spring magnitude, geometry and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_magnitude_geometry_controls.png

Spring full/half-window memory

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_memory_windows.png

Spring increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_increment_tails.png

Spring short/long-lag geometry and circular mixing

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_endpoint_lag_sensitivity.png

Spring largest sampled increment native context

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_large_increment_context.png

All chemical models including Goryachev

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_stochastic_revision.png

Supplementary material and supporting analyses · continued

Chemical memory: both strengths

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_both_strengths_memory.png

Chemical magnitude and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_magnitude_controls.png

Chemical increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_increment_tails.png

Chemical site-gate sensitivity

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_site_gate_sensitivity.png

Original chemical medoids, all18 conditions

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_selected_histories.png

Goryachev lowest-resultant diagnostic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_lowest_resultant_diagnostic.png

Preserved complete mechanism and assay checkpoint

The preceding outline retains all-five mechanism-to-measurement sheets, ring gain/recovery and finite-cue maps, and noise magnitude, controls and full/first/last-window diagnostics. Its numerical evidence is preserved; the revised main-figure claims are stated in this edition.

manuscript_outline/outline_v2.pdf

Prior all-five mechanism evidence crosswalk

Prior complete ring and noise checkpoint

Prior physical motion, magnitude and coverage evidence

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification